

BIOL-1 Practice Test 02 — Answer Sheet

Exam 02 Preparation (Modules 7–11)

Name:		Date:		Score:	
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 **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		10.		19.	
2.		11.		20.	
3.		12.		21.	
4.		13.		22.	
5.		14.		23.	
6.		15.		24.	
7.		16.		25.	
8.		17.			
9.		18.			

Part B: Fill in the Blank (10 questions)

1.		6.	
2.		7.	
3.		8.	
4.		9.	
5.		10.	

BIOL-1 Practice Test 02 — Part C: Short Answer (4 questions)

Free response. Write clearly in the lined spaces.

1. **(Module 7)** Write the complementary DNA sequence AND the transcribed mRNA sequence for the following DNA template strand: **3'-TAC GGG TTA ACG ACT-5'**.
2. **(Module 8 & 9)** Explain the difference between Mitosis and Meiosis in terms of their overall purpose for a multicellular organism like a human. Briefly link how Meiosis connects to Mendel's Law of Segregation.
3. **(Module 10)** Describe how it is possible for a muscle cell and a nerve cell in your body to look and function so differently, despite containing the exact same DNA.
4. **(Module 11)** Choose ONE biotechnology tool discussed in class (e.g., PCR, Gel Electrophoresis, CRISPR, Recombinant Plasmids). Explain what it does and provide one real-world application or example of its use.

Response 1 — Question Number: _____

Response 2 — Question Number: _____

Response 3 — Question Number: _____

Response 4 — Question Number: _____

BIOL-1 Practice Test 02 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 7: Molecular Genetics

1. The enzyme that unwinds the DNA double helix during replication is:
 - A) RNA polymerase
 - B) DNA polymerase
 - C) Helicase
 - D) Ligase
2. Okazaki fragments are synthesized on the:
 - A) Lagging strand
 - B) mRNA transcript
 - C) Leading strand
 - D) tRNA molecule
3. Transcription is the process of converting:
 - A) RNA into a protein
 - B) DNA into RNA
 - C) DNA into a new DNA molecule
 - D) Protein into an observable trait
4. Which part of a primary mRNA transcript is removed during RNA splicing before it leaves the nucleus?
 - A) Introns
 - B) Promoters
 - C) Codons

- D) Exons

5. A mutation that shifts the entire reading frame of the ribosome is called a:

- A) Missense mutation
- B) Silent mutation
- C) Frameshift mutation
- D) Point mutation

Module 8: Cellular Genetics (Mitosis & Meiosis)

1. Which phase of the cell cycle is characterized by DNA replication?

- A) G2 phase
- B) M phase
- C) S phase
- D) G1 phase

2. After a single human cell undergoes meiosis, the resulting cells are:

- A) 4 haploid, genetically distinct cells
- B) 2 diploid, genetically identical cells
- C) 4 diploid, genetically distinct cells
- D) 2 haploid, genetically identical cells

3. Crossing over, which increases genetic diversity, occurs during:

- A) Metaphase II of Meiosis
- B) Prophase of Mitosis
- C) Prophase I of Meiosis
- D) Anaphase I of Meiosis

4. Nondisjunction during meiosis can lead to conditions such as Down syndrome, which is caused by trisomy of chromosome:

- A) 18
- B) 21

- C) 23
- D) Y

5. The p53 protein is heavily involved in:

- A) Forming the cleavage furrow
- B) Breaking down the nuclear envelope
- C) Stopping the cell cycle if DNA is damaged
- D) Attaching spindle fibers to kinetochores

Module 9: Inheritance Genetics

1. The physical appearance or observable trait of an organism is its:

- A) Genotype
- B) Phenotype
- C) Karyotype
- D) Allele

2. In a monohybrid cross between two heterozygous parents ($Aa \times Aa$), what is the expected phenotypic ratio for a completely dominant trait?

- A) 1:2:1
- B) 3:1
- C) 9:3:3:1
- D) 1:1

3. A true-breeding red snapdragon is crossed with a true-breeding white snapdragon, producing offspring that are all pink. This is an example of:

- A) Incomplete dominance
- B) Polygenic inheritance
- C) Sex-linked inheritance
- D) Codominance

4. Human blood types exhibit which type(s) of inheritance pattern(s)?

- A) Sex-linked recessive
- B) Simple dominance only
- C) Incomplete dominance
- D) Codominance and multiple alleles

5. Why are sex-linked recessive traits more common in human males than females?

- A) Males have a stronger Y chromosome.
- B) The X chromosome in males is larger than in females.
- C) Males only have one X chromosome, so a single recessive allele determines the phenotype.
- D) Females lack the genes required for these traits entirely.

Module 10: Epigenetics

1. Epigenetics refers to changes in gene expression that:

- A) Only occur in prokaryotes.
- B) Do not change the DNA sequence but can be passed to daughter cells.
- C) Alter the underlying DNA sequence permanently.
- D) Are only caused by environmental toxins.

2. DNA methylation generally has what effect on gene expression?

- A) It increases transcription.
- B) It causes the DNA to mutate rapidly.
- C) It has no effect on transcription.
- D) It decreases or silences transcription.

3. The lac operon in *E. coli* allows the bacteria to produce enzymes to digest lactose only when lactose is present. This is an example of:

- A) A sex-linked trait
- B) A frameshift mutation
- C) Epigenetic inheritance

- D) Gene regulation

4. Acetylation of histones usually results in:

- A) Loosely packed chromatin and increased transcription.
- B) Immediate degradation of the DNA.
- C) Tightly packed chromatin and decreased transcription.
- D) The splicing of introns out of mRNA.

5. The random inactivation of one X chromosome in female mammalian cells results in physical traits like tortoiseshell coloring in cats. The inactivated X chromosome condenses into a structure known as a:

- A) Centrosome
- B) Barr body
- C) Telomere
- D) Nucleosome

Module 11: Genomics & Biotechnology

1. The primary function of restriction enzymes in biotechnology is to:

- A) Amplify DNA sequences.
- B) Analyze protein folding.
- C) Glue Okazaki fragments together.
- D) Cut DNA at highly specific sequence sites.

2. Polymerase Chain Reaction (PCR) is used to:

- A) Rapidly make millions of copies of a specific DNA segment.
- B) Translate mRNA into a polypeptide chain in a test tube.
- C) Separate DNA fragments by size.
- D) Insert foreign DNA into a bacterial plasmid.

3. During gel electrophoresis, DNA fragments move toward the positive electrode because:

- A) DNA is positively charged.

- B) DNA has a neutral charge but is pulled by gravity.
 - C) agarose gel attracts DNA chemically.
 - D) the phosphate backbone gives DNA a negative charge.
4. The CRISPR-Cas9 system was originally discovered in:
- A) Bacteria as a rudimentary cellular immune system against viral bacteriophages.
 - B) Plants to resist fungal infections.
 - C) Human liver cells to repair damaged DNA.
 - D) Viruses as an attack mechanism against hosts.
5. Transgenic organisms, such as bacteria engineered to produce human insulin, are possible because:
- A) Transcription only works in eukaryotic laboratory cells.
 - B) Bacteria and humans have identical DNA sequences.
 - C) Human genes naturally mutate into bacterial genes.
 - D) All organisms share a nearly universal genetic code.
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Part B: Fill in the Blank (10 questions)

Word Bank: Biotechnology, Cytokinesis, Epigenetic, Heterozygous, Lethal, mRNA, Operon, Smaller, Trisomy, Uracil

1. In RNA, the nitrogenous base _____ replaces the Thymine found in DNA.
2. According to the Central Dogma of molecular biology, DNA is transcribed into _____, which is then translated into a protein.
3. The division of the cytoplasm at the very end of mitosis or meiosis is called _____.
4. The presence of three copies of a chromosome instead of the normal two is referred to as a _____.

5. A gene variant that causes death when inherited in homozygous form is known as a ____ allele.
 6. An individual with two different alleles for a specific trait (e.g., Bb) is called ____.
 7. In gene regulation, a(n) ____ is a cluster of genes under the control of a single promoter, commonly found in prokaryotes.
 8. Identical twins have the exact same DNA sequence, but as they age, differences in their gene expression accumulate due to ____ changes.
 9. In gel electrophoresis, ____ fragments of DNA travel through the gel matrix faster and farther than larger fragments.
 10. The use of an organism, or a component of an organism, to make a product or process for human use is known as ____.
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